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Digital Twin-Based Intelligent Safety Risks Prediction of Prefabricated Construction Hoisting

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Abstract: Prefabricated construction hoisting has one of the highest rates of fatalities and injuries compared to other construction processes, despite technological advancements and implementations of safety initiatives. Current safety risk management frameworks lack tools that are able to process in-situ data efficiently and predict risk in advance, which makes it difficult to guarantee the safety of hoisting. Thus, this article proposed an intelligent safety risk prediction framework of prefabricated construction hoisting. It can predict the hoisting risk in real-time and investigate the spatial-temporal evolution law of the risk. Firstly, the multi-dimensional and multi-scale Digital Twin model is built by collecting the hoisting information. Secondly, a Digital Twin-Support Vector Machine (DT-SVM) algorithm is proposed to process the data stored in the virtual model and collected on the site. A case study of a prefabricated construction project reveals its prediction function and deduces the spatial-temporal evolution law of hoisting risk. The proposed method has made advancements in improving the safety management level of prefabricated hoisting. Moreover, the proposed method is able to identify the deficiencies regarding digital-twin-level control methods, which can be improved towards automatic controls in future studies.

Keywords: Digital Twin; prefabricated construction hoisting; safety risks prediction; intelligent risk prediction

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1. Introduction

Prefabricated construction is a sustainable construction method that enhances productivity and alleviates the adverse environmental effects of conventional construction. Additionally, it can transform the construction mode from traditional manpower to modern industrialization. However, increasing prefabrication rate, component volume, and hoisting height have trigged serious safety issues in hoisting, such as prefabricated components falling, colliding, and hitting [1,2]. These safety problems focus on the hoisting stage of prefabricated components, will cause the fatalities and injuries, and bring enormous losses to construction enterprises. Therefore, the hoisting safety issues should be well solved. In terms of construction safety, the influencing factors of safety accidents are usually found according to the accident reports of on-site construction [3–5]. For example, Liu W et al. [6] adopted a systematic method by exploratory factor analysis (EFA) and structural equation model (SEM) to examine main risk factors by analyzing subway construction accident reports. Meanwhile, the researchers assessed the severity of safety incidents and their impact factors. Raviv et al. [7] used the analytic hierarchy process (AHP) to evaluate risks based on the accident reports of tower cranes. Choe et al. [8] implement the analytic hierarchy process (AHP) to evaluate the quantitative outcome severity level values and compute each event's total risk potential. These methods can effectively sort out the risk events causing safety accidents, and also find out the key influencing factors. However, compared with the post-analysis of safety accidents, the master of

safety risk state in advance can reduce the occurrence of safety accidents in site construction. With the increase in the complexity of prefabricated building projects, the use of scientific and efficient data processing tools to predict construction safety risks has become a new exploration direction.

Artificial intelligence (AI) algorithm can greatly improve the analysis capacity of the massive field data [9], which is more accurate than traditional mathematical models [10,11]. Thereby, using AI algorithms to predict hoisting risks is gaining momentum in current prefabricated building construction safety management studies. For example, Liu M et al. [12] established a safety early warning model for prefabricated building hoisting operations utilizing the Relevance Vector Machine (RVM) algorithm. Liu H et al. [13] provided a prediction model of prefabricated building project risk based on the modified teaching-learning-based-optimization (MTLBO) algorithm. However, AI algorithms are just tools for processing data. To predict the safety risk of hoisting process, it is necessary to build a powerful data interaction system. Therefore, in addition to improving AI algorithms to better handle complex data, it is also important to improve real-time feedback and data acquisition mechanisms.

Digital Twin can integrate various digital means to optimize and improve physical sites through virtual-real mapping, providing a good foundation for safety risks prediction models [14,15]. Digital Twin presents products, processes or entire systems of underlying digital threads in real time through modeling, visualization, prediction and providing feedback on attributes [16]. From the successful application of Digital Twin in the manufacturing industry, it can be seen that the modular construction characteristics of prefabricated building are also suitable for the use of Digital Twin [17,18]. Based on Digital Twin, an overall system of the whole life cycle of serial prefabricated buildings can be built to support disruptive innovation, business flexibility, and a customer-centered construction mode that conforms to circular economy [19]. In terms of construction site management, Digital Twinning can shorten the construction cycle of engineering projects, improve construction quality and construction operation level, and use and recycle resources more effectively throughout the whole life cycle [20,21]. In terms of construction site management, compared to BIM and Internet of Things (IoT) frameworks that have been widely studied, Digital Twin is a management approach that is more closely aligned with on-site changes and has better data analysis capability [22–24]. Therefore, it is valuable to use Digital Twinning to create real-time interactive safety prediction model for assembly building hoisting construction site.

In this paper, an intelligent safety risk prediction framework for the prefabricated hoisting process is proposed. Based on Digital Twin, the framework builds a virtual space model of multi-dimensional simulation mapping of hoisting construction site through the information collected from the site. An improved DT-SVM algorithm is used to accurately predict the safety risk of hoisting. Through the efficient information exchange mechanism and the intuitive and concise visual intelligent safety management platform, the management personnel and field operation personnel can quickly grasp the hoisting safety status information. In addition, a specific case is used to verify the application flow and prediction effect of this Digital Twin framework. This framework provides a high-level safety management method for hoisting construction, which can effectively reduce the casualties and cost loss caused by safety accidents on site.

The rest of the paper is organized as follows: In Section 2, the background and concepts of technologies related to the Digital Twin framework are introduced. Section 3 expounds the whole idea of constructing the Digital Twin framework. In Section 4, the realization process of security risk prediction function of Digital Twin framework is explained. It includes the establishment of virtual—real interaction mechanism and the operation of DT-SVM algorithm. In Section 5, a concrete example is given to verify the prediction effect of the Digital Twin framework and illustrate the operation flow of the framework.

2. Background and Concepts

2.1. Safety Risk of Prefabricated Building Hoisting

At present, the research on the construction safety of prefabricated buildings is gradually enriched, but there are only a few special studies of the hoisting stage. These studies are mainly deployed from two aspects: cause analysis of construction safety accidents and set up information management model [25,26]. In the aspect of the cause analysis of safety accidents, numerical simulation method is usually used to restore the safety accidents of prefabricated buildings, and the management countermeasures are put forward according to the reasons. For example, Roman et al. [27] used a numerical simulation method to study safety accidents in prefabricated building construction and put forward management countermeasures. Liu Z. et al. [28] carried out the numerical simulation of prefabricated components based on BIM technology and analyzed the reliability of prefabricated components. Such numerical simulation methods mainly focus on the post-analysis stage, which is insufficient for accident prevention. Although the complex situation of the accident can be restored to a large extent, it is still challenging to handle the variability and suddenness of the accident. The research on the evolution law of security risk also needs to be deepened.

The safety management model can improve the automation degree of safety management on construction site and provide more efficient guidance. Modeling is a method of system management through simulation and optimization tools. Simulations are used to capture and analyze various details of managed scenarios [29]. Optimizations are used to reduce risks and accidents in managed scenarios [30]. Real-time information interaction between various parts of the model is the key to simulation and optimization operation. At present, most of them rely on integrated IoT model and BIM model to complete [31]. For example, Zhong et al. [32] explored a multi-dimensional BIM platform to improve the real-time visibility and traceability of prefabricated components construction. Gunduz et al. [33] created a dynamic risk management and control model. Lin et al. [34] proposed a structural safety closed-loop management integration framework based on the multi-source data integration. These methods mostly use qualitative means to predict risks in construction. However, the fusion of efficient virtual real interaction methods and algorithms is still a difficult problem to overcome.

2.2. Digital Twin and Machine Learning

2.2.1. Digital Twin

Digital Twin is a key enabling technology to solve the problem of information-physical fusion and to implement the goal of industries intelligence [35]. It has been listed as one of Gartner's top 10 strategic technology trends for four consecutive years (2016–2019) [36]. Professor Michael Grieves of the University of Michigan formally proposed the term Digital Twin in 2003 and it has been widely used ever since [37]. The concept of Digital Twin was first used to solve the maintenance and life prediction problems of aircraft in the future complex service environment. NASA first gave a clear definition of Digital Twin in 2012. Later, under the exploration of scholars, a universal Digital Twin model was gradually formed in the field of intelligent manufacturing. For example, Glatt et al. [38] built an integrated system based on physical simulation, which integrates the actual material handling system and its Digital Twin system. Ding et al. [39] summarized the multi-dimensional and multi-scale intelligent manufacturing space based on Digital Twin and its modeling method. The achievements of Digital Twin in manufacture industry also inspire the development of the construction industry towards intelligent construction. Tao et al. [40] elaborated the relationship among the urban big data, virtual-real interaction, and intelligent services, and then developed the operating framework of the Digital Twin city. Liu et al. [41] proposed an intelligent construction method based on Digital Twin to adjust and amend the actual construction process. Dongmin et al. [42] developed a framework for integrating Digital Twin and blockchain. These methods generally integrated the

internet of things (IoT), building information modeling (BIM) and finite element model to build Digital Twin frameworks. These Digital Twin frameworks achieve almost real-time updates to optimize construction management.

Tao et al. [40] summarized the Digital Twin framework in the fields of vehicles, electromechanical equipment, space communication network, three-dimensional warehouse, medical care, smart city and so on. It makes a lot of contributions to the extension of the application of Digital Twin technology. The theory of Digital Twin is also being applied to the construction industry. In this paper, a framework of Digital Twin in the field of building construction is proposed. This is shown in Figure 1 [24].

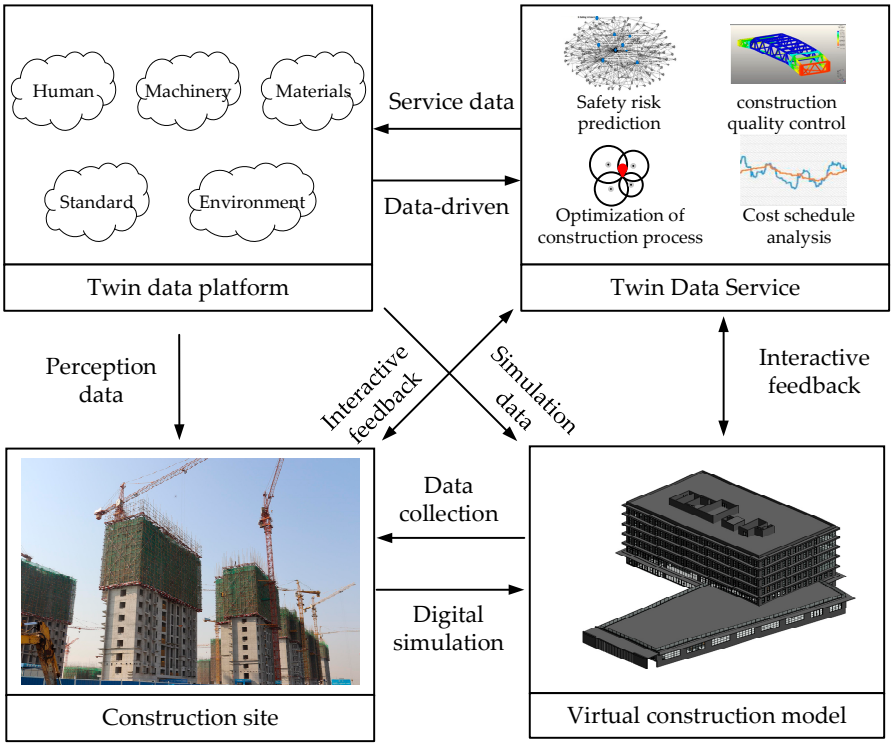


Figure 1. Digital Twin frame in construction field.

Based on the traditional physical construction site, the framework establishes a virtual construction model from four aspects: geometry, physics, behavior, and rules. The data between the physical model and the virtual construction site interact in real-time. At the same time, the physical construction site and virtual construction model data are uploaded to the cloud to form a twin data platform. The twin data platform classifies the data according to the dimensions of human, machine, material, method, and environment. With decision forest, BP neural network, support vector machine, deep belief network, and other algorithms as the driver, the framework is able to achieve cost schedule analysis, safety risk prediction, quality control, construction process optimization, and other services. This framework can improve the information level of the construction process and reduce the risk in the construction.

Although there appears to be some success in current applications, Digital Twin frameworks specifically designed to solve prefabricated hoisting safety problems remain to be developed.

2.2.2. Potential of Machine Learning for Digital Twin in Construction

Machine learning is the key technology of artificial intelligence. It plays a crucial role in big data processing and brings significant value in saving time for safety management and Optimize resource allocation in computation [43]. The fusion and processing of massive data is the key element of the Digital Twin framework. Thus, machine learning is the

core driver of the Digital Twin framework. In recent years, machine learning has also made breakthroughs in image recognition, speech recognition, natural language processing, and other aspects [44]. The application advantages of machine learning make scholars find new methods to solve the construction safety problems in the construction industry. In terms of safety risk identification, Zhang et al. [45] used machine learning to analyze construction accident reports and classify the cause of the accident. Ding et al. [46] proposed a new mixed deep learning model to identify unsafe behaviors in subway construction. Wang et al. [47] proposed new technology for fast and automatic detection of weathering and spalling damage of ancient buildings based on deep learning. Ni et al. [48] conducted a quantitative crack width measurement method based on digital language and dual-scale convolutional neural networks.

In terms of construction safety risk assessment and prediction, the main difficulty associated with this process is the multi-variate and nonlinear relationship between parameters and risk levels [49]. At present, commonly used machine learning methods include artificial neural network (ANN) [50], SVM [51] and Bayesian network (BN) [52], etc. BN can predict the safety risk state through the causal relationship between risk events and accidents, but its structure is highly subjective and requires a large number of sample data to learn [53]. ANN has widespread application in forecasting, but is still a black-box-like system [54]. In contrast, the kernel function of SVM has a solid research foundation and has significant advantages in solving small sample size, high dimension and nonlinear problems [55]. Therefore, SVM is widely used in engineering safety risk prediction. For example, Zhou, C. et al. [56] proposed a hybrid deep learning model based on attitude and position prediction framework for shield tunneling. Zhou, Y., et al. [57] proposed a new safety risk prediction method for deep foundation pit construction of subway infrastructure based on SVM. However, there are few researches on prefabrication construction safety risk management using machine learning, especially in hoisting construction.

3. Digital Twin-Based Prefabricated Building Hoisting Risk Prediction Framework

Combined with the risk factors of prefabricated building hoisting and the Digital Twin framework of the construction process, a multi-dimensional Digital Twin model M_{DZ} for safety risk prediction of prefabricated buildings hoisting is proposed. M_{DZ} is shown in Formula (1):

$$M_{DZ} = (PCS, DVT, DD, SS, CN) \quad (1)$$

where M_{DZ} represents the Digital Twin multi-dimensional model for prefabricated building hoisting. *PCS* stands for the physical hoisting process. *DVT* stands for hoisting virtual model. *DD* stands for big data storage management platform. *SS* stands for hoisting safety risk prediction service. *CN* represents data acquisition and transmission between physical hoisting process, virtual hoisting model, big data management platform, and hoisting risk prediction service.

According to Formula (1), a prefabricated construction hoisting method based on Digital Twin is proposed, as shown in Figure 2. In this method, Digital Twin concept and Internet of Things technology are applied to collect safety risk data (*CN*) of the physical hoisting process (*PCS*). The collected data will be transmitted to the big data storage management platform (*DD*). On the one hand, the platform processes and stores the data. On the other hand, the proposed algorithm is used to analyze the data, and then hoist safety risk prediction (*SS*) is carried out. The virtual construction model (*DVT*) is used as a visual interface to accurately and intuitively express the result of hoisting safety risk prediction. The predicted results as instant feedback will be automatically transmitted into the physical hoisting process (*PCS*) through the actuators and controllers. Through the above process, the real-time interactive feedback between the physical process (*PCS*) and the virtual model (*DVT*) is investigated. The framework finally conducts the service function (SS) of hoisting process safety risk prediction.

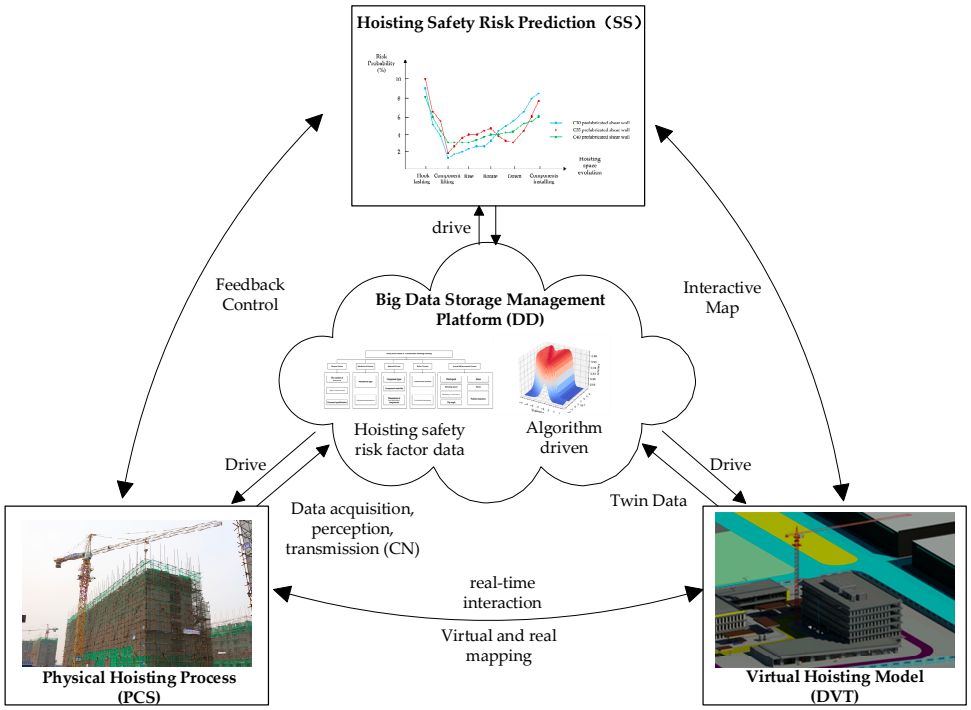


Figure 2. Safety risk prediction framework in hoisting process.

4. Safety Risk Prediction Method for Hoisting Process Based on Digital Twin

4.1. Prefabricated Component Hoisting Safety Risk Identification

Risk identification is the premise of safety risk prediction. First of all, literature [58–65] was consulted to conduct field research on the hoisting process. According to statistics, hoisting accidents mainly include falling objects, object impact, mechanical accidents, component damage and so on. The causes of hoisting accidents include personnel, machinery, materials, methods and environment. In this paper, Work Breakdown structure-risk Based Supervision (WBS-RBS) analysis was used to decompose the hoisting process in terms of time sequence and element extraction. Obtain security risk factors.

4.1.1. WBS-RBS Decomposition of Prefabrication Hoisting Safety Risk Factors

- WBS Decomposition

For the hoisting process, WBS decomposition is carried out according to time. The hoisting process is decomposed according to the time, which can be divided into hoisting stage, hoisting stage and installation stage. The decomposition is further refined, and the hoisting stage can be divided into crane movement, component binding and component trial hoisting. Hoisting stage can be divided into component hoisting, component rotation, component descent; Installation stage can be divided into component fine tuning, component placement, component untying.

- RBS Decomposition

For the hoisting process, RBS decomposition was carried out according to the elements. The hoisting process is decomposed according to the main elements involved, which can be divided into personnel factors, equipment factors, material factors, management factors and environmental factors. The personnel factor can be further divided into the number of personnel, personnel type, personnel qualification. Equipment factors can be divided into mechanical types and mechanical parameters. Material factors can be divided into component types and component materials. Management factors can be divided into construction scheme and technical disclosure. The main environmental factor is wind speed.

4.1.2. WBS-RBS Analysis of Prefabrication Hoisting Safety Risk Factors

As shown in Table 1, the working package W decomposed by WBS in the hoisting process and the working package R decomposed by RBS in the hoisting process. The WBS-RBS matrix was constructed by cross coupling W and S. The risks of matrix elements are judged one by one and comprehensively identified systematically.

Table 1. WBS-RBS decomposition of hoisting.

		W1			W2			W3		
		W11	W12	W13	W21	W22	W23	W31	W32	W33
R1	R11	0	1	0	0	0	0	0	0	0
	R12	0	0	0	0	0	0	0	0	0
	R13	0	1	0	0	0	0	1	0	0
R2	R21	0	0	0	1	1	1	0	0	0
	R22	0	0	0	1	1	1	0	0	0
R3	R31	0	0	0	1	1	1	0	0	0
	R32	0	0	0	1	1	1	0	0	0
R4	R41	0	1	0	0	0	0	0	0	0
	R42	0	1	0	0	0	0	0	0	0
R5	R51	0	0	0	1	1	1	1	0	0

In the hoisting risk WBS-RBS decomposition, "1" is used to represent the coupling unit if there is a risk, and "0" is used to represent the coupling unit if there is no risk. According to the decomposition results of hoisting risk WBS-RBS, two obvious aggregation phenomena can be found. One is the local aggregation of W21, W22, W23 and R21, R22, R31, R32. The other is the linear focusing of R51.

According to the above analysis, the logical diagram of the risk causes in the hoisting process is shown in Figure 3.

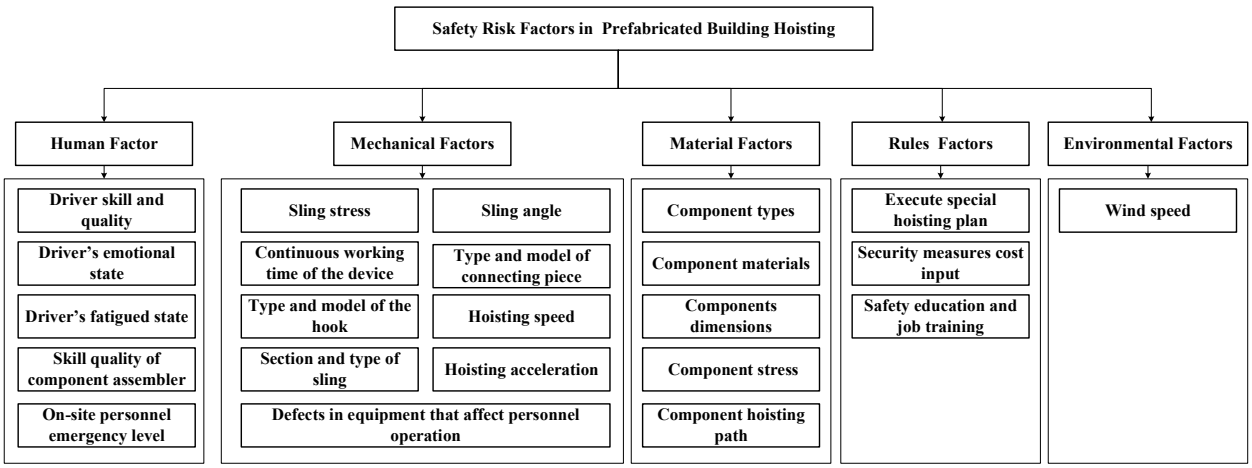


Figure 3. Logic diagram of safety risk in hoisting process.

4.2. Collection and Transmission of Hoisting Information

In the safety risk prediction framework of the hoisting process, a Digital Twin model corresponding to the physical construction site should be established. To analyze the causes of hoisting safety risks, the Digital Twin model needs to characterize the relevant information of hoisting safety risk prediction.

The premise of safety risk prediction in the hoisting process is data collection. Digital Twin can be used to obtain the multi-dimensional data of construction site. As shown in Figure 4, the Digital Twin data acquisition configuration mainly includes three aspects:

intelligent component configuration (*ICC*), intelligent device configuration (*IDC*), and intelligent gateway configuration (*IGC*). The following will elaborate on these three aspects.

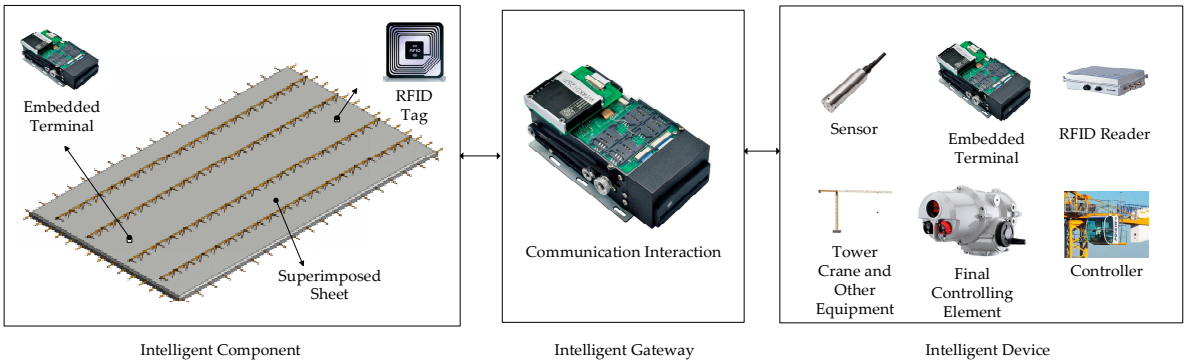


Figure 4. Digital Twin data acquisition and transmission in hoisting process.

- Intelligent component configuration (*ICC)*

ICC is composed of component, active RFID tag, sensor, and embedded terminal. In the process of component manufacturing, the active RFID tag and embedded terminal are closely attached to the component. Among them, the component itself refers to the prefabricated components such as laminated panels, stairs, walls, balcony slabs, and columns. Active RFID tags are used for component identification and component information writing. The unique identification ID, type, material, geometric size, storage location, and other information of the component are stored in the tag. By indexing component ID, the construction technology of components and all the quality requirements of each construction process can be queried in the cloud database. The sensors are mainly strain sensors [55] and positioning sensors. They can sense changes in mechanical properties and position of components in real-time and are connected to embedded terminals. The embedded terminal has the capability of data acquisition and transmission. It can transmit the data collected by sensors to virtual models and cloud processors in real-time.

- Intelligent device configuration (*IDC*)

The *IDC* consists of the device, the embedded terminal, the RFID reader, the heterogeneous sensor, the actuator, and the controller. Among them, the equipment itself mainly refers to the tower crane. RFID readers are used to record the state of the component and write active RFID tags. It can communicate wirelessly with active RFID tags of smart components. Heterogeneous sensors mainly perceive data such as position data, speed data, and stress data. Actuators and controllers are used to receive execution instructions and execute control commands. Workers on the construction site are also included in the category of generalized actuators. The embedded terminal has the capability of data acquisition and transmission and can upload multi-dimensional data collected by the sensor to the cloud server. Similar to the embedded terminal in *ICC*, it uploads multi-dimensional data collected by sensors to the cloud server and sends the data processing results from the cloud to the actuator as instructions.

- Intelligent gateway configuration (*IGC*)

IGC is the part that builds the connection between the sensor and Long Range Radio (LoRa) embedded terminal. Intelligent gateway is based on LoRa technology and arranged in star network architecture. First, sensors placed on the components and devices feed the collected real-time information to the LoRa embedded terminal for integration. Then, the communication module in the LoRa terminal is used to complete the information transmission, and its transmission architecture is shown in Figure 5. Information data can be transmitted to Microsoft Access public database in XML format through 4G network. The information can also be transmitted to Microsoft Access public database in

real-time through the gateway in the form of field AD hoc network. Local servers and client terminals, as well as PC and other devices, can access the common database in real-time to obtain the information during the hoisting process.

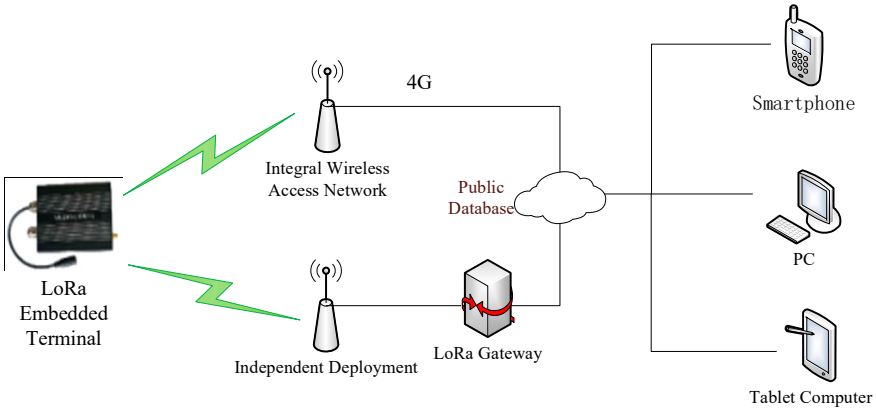


Figure 5. LoRa Network Information Transmission Architecture.

- Identification and storage structure of prefabricated component information

The component status is located and monitored by sensors on site. Workers scan RFID tags to confirm whether the hook is installed and whether the components are installed in the building structure. When the prefabricated component is installed on the hook, the pre-hoist component's information and the installed status of the hook are confirmed by the worker scanning. Meanwhile, the data center starts to receive and update sensor data. These data are stored in the component's information dataset. The storage structure of each prefabricated component information data set in the Access database is shown in Figure 6. After scanning the RFID tag, the data center will analyze the stress sensor status of the hook and the component hoisting point to ensure that their stress is qualified. The data center delivers data to intelligent algorithm models and visualization models for prediction and data visualization. When the prefabricated component arrives at the installation site, the sensor data gradually regress to the state out of the hook. The worker scans the RFID tag to confirm that the component is installed properly. The data center stores the component's dataset and finishes updating it.

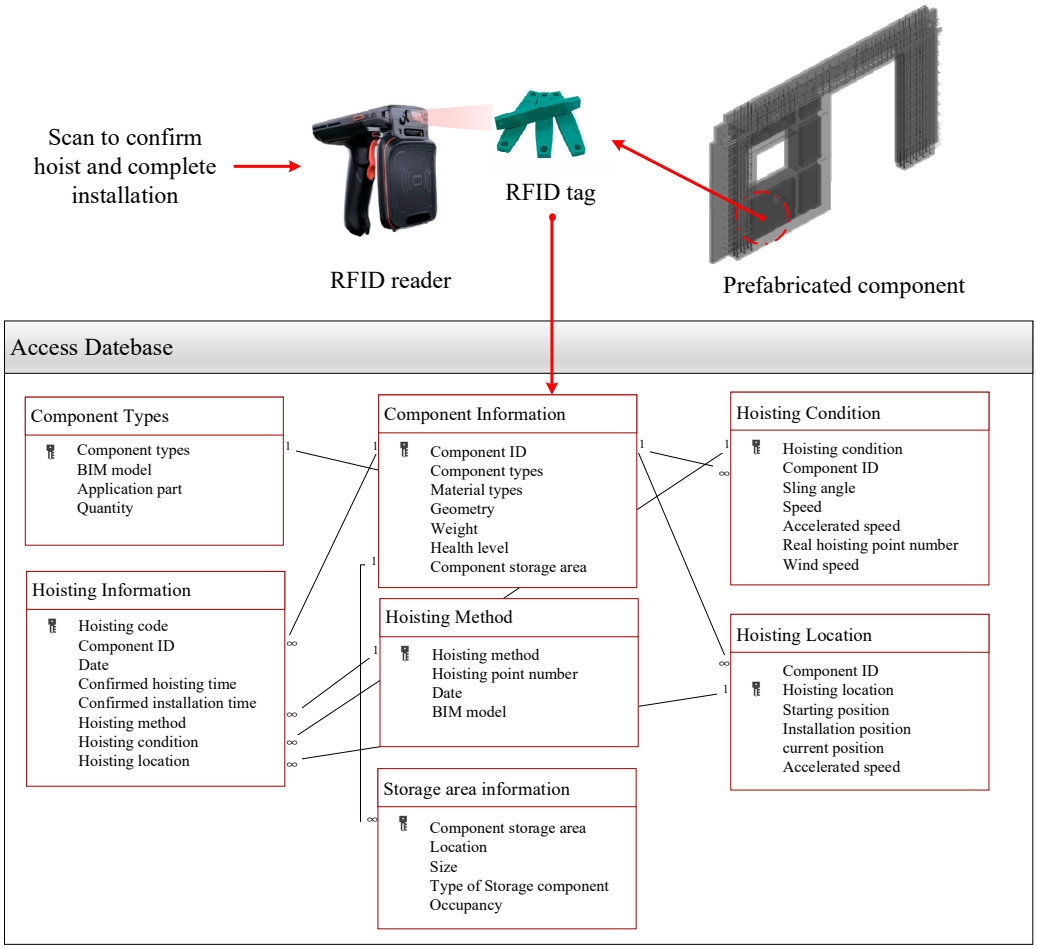


Figure 6. The storage structure of each prefabricated component information data set in the Access database.

4.3. Hoisting Risk Prediction Based on Multi-Source Heterogeneous Data

4.3.1. DT-SVM Algorithm Based on Digital Twin Multi-Source Heterogeneous Data

1. Normalization of sample data;

Through the analysis of the prefabricated component hoisting safety accident, the probability of the accident is positively correlated with the stress of the sling and the component. The occurrence of hoisting accident depends on many factors, such as sling angle, hoisting speed, hoisting acceleration and so on. Therefore, the functional relationship to determine the stress of sling and component is put forward as Formula (2).

$$Ss, Cs = f(Sa, Hv, Ha, Ws, Hm, Ct, Cm, Cd) \quad (2)$$

where *Ss*, *Cs*, *Sa*, *Hv*, *Ha*, *Ws*, *Hm*, *Ct*, *Cm*, and *Cd* are sling and component stress, sling angle, hoisting speed, hoisting acceleration, wind speed, hoisting method, component type, component material and dimension of prefabricated component.

When data variables in the relational function are collected, the field dataset is defined as U'. Due to the combinatorial nature of logical and numerical data, the collected datasets U' cannot be directly applied to the algorithm operation. Therefore, the derivation of a normalized sample dataset U from the original data is of essence to predict the hoisting safety risk.

U' is divided into continuous numerical dataset A' and discrete logical dataset B'.

Continuous numerical dataset A_i' mainly includes continuous working time of equipment, sling stress, hoisting speed, component stress, wind speed, etc. Where represents the number of different risk factor units. The continued data unit a_{ik}' is all numerical in form but with different dimensions. To ensure the operational efficiency of machine learning, numerical data with different dimensions are normalized to generate a data unit a_{ik}. Integrate a_{ik} into the dataset A_i. The integration process is shown in Formulas (3) to (5):

$$A_i' = [a_{i1}', a_{i2}', \ldots a_{ik}', \ldots, a_{in}'] \tag{3}$$

$$f: a_{ik}' \to a_{ik} = \frac{a_{ik}' - a_{\min}'}{a_{\max}' - a_{\min}'} \tag{4}$$

$$A_i = [a_{i1}, a_{i2}, \ldots a_{ik}, \ldots, a_{in}] \tag{5}$$

Discrete logical data B_j' mainly includes the skills and qualities of drivers, the emotional state of drivers, the skills, and qualities of component assemblers, the implementation of special hoisting schemes, etc. Discrete data unit b_{jk}' is a qualitative variable and stored with text information. It cannot be directly applied to machine learning operations and needs to be quantified. Firstly, the safety risk factors corresponding to the discrete data can be classified and processed according to engineering experience and divided into different sets C_n. Secondly, the discrete data set need to transform into a quantitative numerical unit b_{jk} through piecewise function operation. Thirdly, the quantitative numerical unit b_{jk} is integrated into a dataset B_j. The integration process is shown in Formulas (6) to (8):

$$B_j' = [b_{j1}', b_{j2}', \ldots b_{jk}', \ldots, b_{jn}'] \tag{6}$$

$$f: b_{jk}' \to b_{jk} = \begin{cases} 1(b_{jk}' \in C_1) \\ 2(b_{jk}' \in C_2) \\ 3(b_{jk}' \in C_3) \\ \vdots \\ d(b_{jk}' \in C_d) \end{cases} \tag{7}$$

$$B_j = [b_{j1}, b_{j2}, \ldots b_{jk}, \ldots, b_{jn}] \tag{8}$$

2. Conversion of multi-source isomorphic data to uniform sample data

Data unit U' is the union of data set *A* and *B*. Collections *A* and *B* are stored in the same matrix as row vectors arranged from top to bottom, each row representing different data source. The multi-source row vector unit $A_i(B_j)$ was transformed into column vector sample unit u_r by matrix transformation. Integrate u_r into uniform isomorphic sample matrix U. Column vector unit u_r is a sample containing all risk factor characteristics. The transformation process is shown in Formula (9):

$$U' = \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_i \\ \vdots \\ A_p \\ B_1 \\ B_2 \\ \vdots \\ B_j \\ \vdots \\ B_q \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2k} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ik} & \cdots & a_{im} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pk} & \cdots & a_{pm} \\ b_{11} & b_{12} & \cdots & b_{1k} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2k} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{j1} & b_{j2} & \cdots & b_{jk} & \cdots & b_{jm} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{q1} & b_{q2} & \cdots & b_{qk} & \cdots & b_{qm} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{21} & \cdots & u_{k1} & \cdots & u_{m1} \\ u_{12} & u_{22} & \cdots & u_{k2} & \cdots & u_{m2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ u_{1r} & u_{2r} & \cdots & u_{kr} & \cdots & u_{mr} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ u_{1s} & u_{2s} & \cdots & u_{ks} & \cdots & u_{ms} \end{bmatrix} = [u_1, u_2, \cdots, u_m] = U \quad (9)$$

Then, the safety state of these datasets is calculated according to whether the stress of sling and component meets 80% of their respective limit stress as the judgment criterion [56,57]. In total, 80% of these data sets are used for model training and 20% for model testing. Additionally, the sample data sets need to be randomly shuffled.

3. Generation of decision function.

Unified sample data U is imported. To optimize the model, the kernel function and penalty parameters C are selected, and the optimal solution $\alpha^* = \left(\alpha_1^*, \cdots, \alpha_l^*\right)^T$ is constructed and obtained. Compared with polynomial kernel function, radial kernel function has fewer parameters, which can reduce the complexity of SVM model and realize nonlinear mapping. The DT-SVM function contains the support vector U and the inner product (x_i, x) of the training sample. The inner product kernel function $k(x_i, x)$ is used to replace them. The original eigenspace is transformed into a new high linear eigenspace by the nonlinear transformation of the inner product kernel function. Select a positive component α_j^* of α^* less than C, set tolerance deviation ε, define the edge of the "interval band", and calculate ω^* and b^* accordingly. To relax the interval requirement of the function, the relaxation variable ξ_m is introduced to optimize the function. The decision function $f(x)$ is as shown in Formula (10):

$$f(x) = \sum_{i=1}^{l} (\alpha_j^* - \alpha) k(x_i, x) + b^* \quad (10)$$

where x_i corresponding to $\alpha_j^* - \alpha \neq 0$ is the support vector; the decision function $f(x)$ is the rule that divides the training sample into two parts. That is, centering on $f(x)$, an interval belt with a width of 2ε is constructed. Whether the training samples fall into this interval is taken as the boundary to distinguish whether there is a hoisting safety risk.

4. Evaluation of prediction effect.

In addition, the correlation coefficient R^2 is used to evaluate the prediction effect. The closer the value R^2 is to 1, the more accurate the prediction result is. The correlation coefficient R^2 is calculated by the Formula (11):

$$R^2 = \frac{\sum_i (y_i - f_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (11)$$

where y_i represents the actual value; f_i represents the predicted value; $\overline{y}$ represents the average value of the actual value. When the accuracy is not high, the model needs to be retrained.

4.3.2. Hoisting Safety Risk Prediction Process Based on DT-SVM

As shown in Figure 7, for multi-source heterogeneous hoisting data, the specific steps of SVM-based hoisting safety risk prediction mainly include:

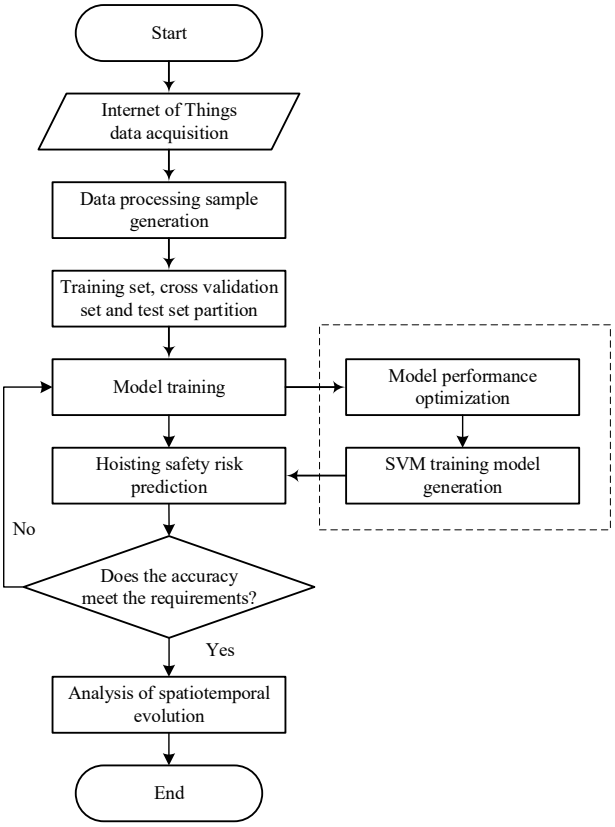


Figure 7. Hoisting safety risk prediction process based on DT-SVM.

1. Normalization of sample data. The sample data are derived from the Digital Twin model in the hoisting process. Data types include numeric and logical data. The logical data needs to be converted into numerical data. The dimension difference of the influence factor will lead to the significant difference of the value. Therefore, normalization is needed, and deviation standardization is the most commonly used method.
2. Selection of training set and test set. To prevent overfitting, at least 80% of the sample data are used for model training. The rest of the sample data are used for testing, and the sample data need to be randomly scrambled. DT-SVM prediction requires that the corresponding classification label should be set for the risk level. Since categorical labels do not affect DT-SVM classification performance, they can be set by themselves.
3. Model training. The key to data training is to choose the proper kernel function and parameters. The radial basis kernel function (RBF) is superior to other kernel functions in both accuracy and computational performance. The penalty coefficient C and kernel parameter γ are usually determined by k -fold cross-validation.
4. Evaluation of testing and effectiveness. The test is a calculation of a selected set of tests. The safety risk level was obtained through the prediction model, and the correlation coefficient R^2 was calculated by comparing it with the actual risk level. The

closer the value R^2 is to 1, the more accurate the prediction result is. If the value of R^2 indicates a low accuracy, the above steps 3. and 4. should be repeated.

5. Analysis of space-time evolution. After the prediction model is established, the collected and sorted data are successively imported into the prediction model. After calculation, the risk states of different periods and positions in the hoisting process are obtained. According to the state of each risk indicator predicted by DT-SVM model, the probability of each indicator is calculated. A working condition is formed by combining the indicators at a certain moment in turn. The working condition with a probability of more than 70% is set as a dangerous working condition. The ratio of dangerous working conditions to all working conditions is the risk probability at this moment. The temporal and spatial evolution of risk was analyzed according to risk probability.

4.4. Construction and Operation of Virtual Model of Hoisting Process

To intuitively display the hoisting process status and safety risk prediction results to managers, a virtual model needs to be built. The virtual model needs to have three functions: First, it can visually display the hoisting process. Second, the real-time display of hoisting risk. Third, the risk of the issue of control instructions for control. The above functions can be conducted through three steps: BIM model processing, information and database link, and feedback control.

1. Processed BIM model. Firstly, build the lightweight model. The BIM model is generated into an IFC format file, which is parsed and read under the JavaScript environment. Then use WebGL technology to carry the BIM model on the web page.
2. Link the information to the database. By carrying the original information of sensors in the public database into the BIM model, the location of sensors and the collected real-time data can be viewed on the webpage. At the same time, the virtual model can simulate the physical construction hoisting site according to the information of the positioning sensor. This simulation has a corresponding relationship in time and space, and is able to transform the traditional 3D BIM model into a 4D model based on time dimension under the Digital Twin.
3. Feedback control. When there is a safety risk in the forecast result of hoisting, the system will issue an alert. There are two ways to alert. One way is to highlight the time node and risk position of the risk on the web page. The other part is to link the alarm device through the LoRa system and carry out flash sounds at the corresponding place on the construction site. At the same time, the hoisting risk diagnosis report is transmitted to the operator in real-time through APP.

5. Case Study

5.1. Project Background

The proposed framework is applied to a large prefabricated project with a total construction area of 71,172.43 square meters as the case study. The structure type is reinforced concrete shear wall structure. The assembly rate of this project is 37.5%, the maximum height is 54 m, and the highest floor is 18 floors. The parts of prefabricated components used in prefabricated engineering are prefabricated composite floor slabs, prefabricated stair slabs, and partial prefabricated shear walls. The construction site is shown in Figure 8. The prefabricated construction method and prefabricated components used in this project are universal. Therefore, this case is representative to illustrate the operation process of the prediction method.

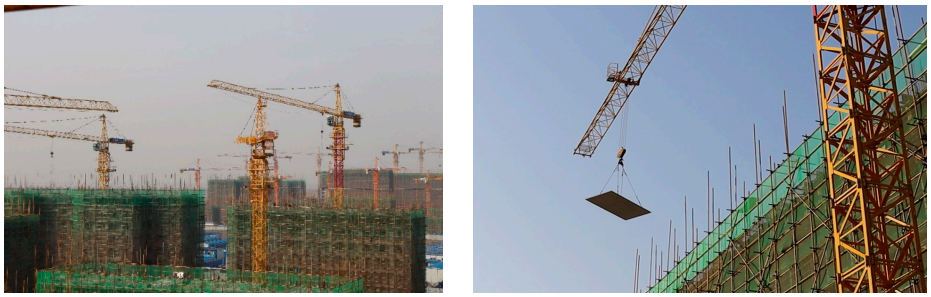


Figure 8. Prefabricated project construction site.

The case will take the hoisting of the composite slab as the main research object. The whole DT framework starts from five aspects: human, machine, material, law and environment. The prediction method is not specific to a certain type of component or accident type and is generally applicable to prefabricated component hoisting. Firstly, sensors are arranged to collect data, which will be transmitted to the cloud database through the embedded terminal. According to the data information, the multi-dimensional and multi-scale hoisting virtual model is established. The DT-SVM algorithm is applied to predict the hoisting safety risk of multi-source data stored in the cloud. The temporal and spatial evolution law of hoisting safety risk is analyzed, and the corresponding risk control method is given.

5.2. Hoisting Data Acquisition and Transmission

The collection and transmission of hoisting data is the basis of safety risk prediction. Management factors are associated with significant uncertainties such as man, method, environments, and so on. Therefore, on the basis of consulting relevant experts, this paper selects eight parameters as characteristic variables: sling angle, hoisting speed, hoisting acceleration, wind speed, hoisted method, component types, component materials, and dimensions of prefabricated components. In addition, the stress of sling and component is the main factor affecting the safety of hoisting. Their monitoring values are also used as a measure of the risk of hoisting. When making risk predictions, relevant data of the above eight variables need to be collected.

The data related to the above factors can be divided into static data and dynamic data. The static data include component material, component type, number of hoisting points, hoisting method, crane model, the position of hoisting points, and inclination angle of hoisting rope. Static data can be captured through the BIM model. Dynamic data include hoisting acceleration, hoisting speed, wind speed, sling stress, and component stress. Dynamic data changes fast, so it is necessary to build the Internet of Things (IoT) to dynamically acquire data.

First, lay out the sensors. The types of sensors include positioning sensors, acceleration sensors, wind speed sensors, strain sensors, and inclination sensors. After the sensor layout is completed, the intelligent gateway needs to be arranged. Intelligent gateway is based on LoRa technology and is arranged in a star network architecture to build the connection between sensors and LoRa embedded terminals. Real-time information collected by sensors that are placed on components and devices is first fed to the LoRa-embedded terminal for integration. Then the communication module in the LoRa terminal is used to transmit the data to Microsoft Access public database in XML format through the 4G network. The local server and the client acquire the information during the hoisting process by accessing the public database. The database conducts the record and the statistics to the data.

The hoisting data are collected and stored in the database. The construction operators can view the original data of the hoisting process through the hoisting data collection module. The data will be divided into different hoisting stages according to the value of velocity and acceleration. In the Start phase, the initial velocity is 0 and the acceleration is

positive. In the deceleration phase, the initial velocity reaches the maximum and the acceleration is negative. In the rotation phase, both velocity and acceleration are 0. In the descending stage and the deceleration stage before the end, the acceleration and velocity are opposite to the lifting stage. The records of logical characteristic variables are represented by codes, which refer to Table 2. The original data collected in the hoisting process is shown in Figure 9, and the first 10 records are selected as examples of data collection in each stage.

Table 2. Division of influencing factors.

Characteristic Variable	Data Type	Maximum Value	Minimum Value	Average Value	Mean Square Error
Sling angle A_1/°	Numeric	59.99	30.06	44.85	8.65
Hoisting speed A_2/m/min	Numeric	79.93	−79.68	41.02	46.81
Hoisting acceleration A_3/m/s^2	Numeric	0.045	−0.045	0.02	0.03
Wind speed A_4/m/s	Numeric	4.98	2.04	3.51	0.86
Hoisted method B_1	Logical	1-four points hoisting, 2-six points hoisting			
Component types B_2	Logical	1-prefabricated wallboard, 2-composite floor slabs			
Component materials B_3	Logical	1-C30 reinforced concrete; 2-C35 reinforced concrete; 3-C40 reinforced concrete			
Dimension of prefabricated component B_4	Logical	1-2750 × 1500 × 200 (wallboard); 2-180 × 3320 × 60 (floor)			

Unclassified

		1	2	3	4	5	6	7	8
Number	State	Hoisting speed	Hoisting acceleration	Hoisted method	Component types	Dimension of prefabricated	Component materials	Sling angle	Wind speed
1	Start	0	0.045	1	1	1	1	30	4.1011
2		0	0.043	1	1	1	1	30	3.913
3		0	0.045	1	1	1	1	30	4.1233
169	Decelerate (1)	80	-0.044	1	1	1	1	30	2.7067
170		80	-0.045	1	1	1	1	30	3.8105
171		80	-0.045	1	1	1	1	30	4.0542
337	Rotation	0	0	1	1	1	1	30	2.9177
338		0	0	1	1	1	1	30	3.2368
339		0	0	1	1	1	1	30	4.3971
505	Down	0	-0.044	1	1	1	1	30	3.294
506		0	-0.045	1	1	1	1	30	3.8397
507		0	-0.045	1	1	1	1	30	2.9142
673	Decelerate (2)	-80	0.045	1	1	1	1	30	2.9166
674		-80	0.043	1	1	1	1	30	2.883
675		-80	0.045	1	1	1	1	30	2.4632
841	End	0	0	1	1	1	1	30	2.2483
842		0	0	1	1	1	1	30	4.7108
843		0	0	1	1	1	1	30	2.8076

Figure 9. Hoisting data collection.

5.3. Hoisting Safety Risk Prediction and Analysis Based on Multi-Source Heterogeneous Data

5.3.1. Hoisting Safety Risk Prediction Based on DT-SVM Algorithm

1. Normalization of sample data;

The above datasets are formed into multi-source heterogeneous data, and the influencing factors are divided as shown in Table 2.

This case mainly aims at two accidents of falling objects and damage of components. However, the falling objects and component damage are served as the illustrative scenarios to demonstrate and validate the applicability of the proposed framework. The proposed DT framework in this paper is general and can be applied to other types of accident scenarios.

When an accident occurs, the most significant features that can be detected are changes in the stress of the sling and members. In this project, the total breaking tension of sling was 31.3 KN. The ultimate stress of concrete C30, C35, C40 is 2.01 N/mm^2, 2.20 N/mm^2, 2.39 N/mm^2, respectively. To maintain a sufficient safety margin, 80% of the ultimate stress is taken as the limit. If the stress of the sling and the laminated plate exceeds

80% of their corresponding ultimate stress, then the hoisting process is considered potentially hazardous; otherwise, the sample is deemed safe.

Numerical data with different dimensions were normalized to generate the data unit a_{ik}, which was integrated to form numerical continuous dataset A_i. The discrete dataset, on the other hand, was transformed into the quantitative numerical unit b_{jk}, and finally integrated into the dataset B_j. The matrix transformation was performed on the multi-source row vector unit $A_i(B_j)$. Then the transformed column vector sample unit was integrated into a sample matrix U. The integration process is shown in Formula (12). The column vector unit u_r is a sample containing all risk factor characteristics. Thus, the unified sample data of hoisting process could be generated for machine learning operation.

$$U' = \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ B_1 \\ B_2 \\ B_3 \\ B_4 \end{bmatrix} = \begin{bmatrix} -0.495 & -0.328 & \cdots & 0.172 & \cdots & 0.505 \\ -0.256 & 0.243 & \cdots & -0.756 & \cdots & -0.256 \\ 0.278 & -0.722 & \cdots & -0.222 & \cdots & 0.277 \\ 0.201 & -0.197 & \cdots & 0.441 & \cdots & 0.248 \\ 1 & 1 & \cdots & 2 & \cdots & 2 \\ 1 & 1 & \cdots & 2 & \cdots & 2 \\ 1 & 1 & \cdots & 2 & \cdots & 3 \\ 1 & 1 & \cdots & 2 & \cdots & 2 \end{bmatrix} = [u_1, u_2, \cdots, u_r] = U \quad (12)$$

2. Establishment and optimization of safety risk prediction model;

The data of 130 samples were used as the data source. Radial kernel function (RBF) was selected as the modeling function. Penalty function *C* and kernel function *G* have significant effects on model performance. Therefore, the k-fold cross-validation method is used to select the optimal parameters of the penalty function *C* and kernel function γ. Digital Twin feedback requires real-time performance. Therefore, when selecting parameters, model accuracy is taken as the main measurement standard, and operation time is taken as the supplementary measurement standard. The accuracy and calculation time of DT-SVM algorithm in selecting different penalty function C and kernel function γ are shown in Figure 9.

lg γ and lg*C* were taken as X and Y axes, respectively, and the accuracy of the model was taken as Z axis to establish a 3D coordinate system. It can be seen that when lg*C* was within a certain range, the accuracy of the model was at a high level. The highest accuracy points of the model (0.53, 2.55, 99.3) were obtained, and the corresponding lg*C* and lg γ were 0.53 and 2.55, respectively. The optimal parameters of the penalty function *C* and kernel function γ can be obtained. The penalty function *C* was 0.435, and the optimal parameter was 0.250. At this time, the accuracy of the model reached 99.3%, and the calculation time was 6.685 s. The calculated value of R^2 was greater than 0.89, which is close to 1. This indicated that the prediction results of assembly hoisting safety risks of this DT-SVM model are more accurate.

5.3.2. Analysis on the Spatial-Temporal Evolution of Hoisting Safety Risk

To study the spatial-temporal evolution of hoisting safety risks, researchers collected a large amount of data during the composite plates hoisting process. The data come from experimental data and simulation data. The data were input into the above established DT-SVM risk prediction model for processing. The temporal and spatial evolution law of safety risk in hoisting process was obtained. As shown in Figure 10, the predicted results were basically consistent with the actual situation.

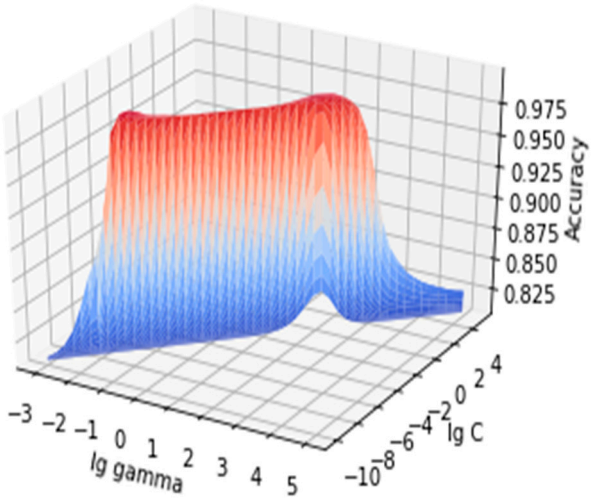


Figure 10. Three-dimensional function diagram of cross-validation accuracy of different penalty functions *C* and kernel functions *G*.

According to the line chart of the evolution law of hoisting safety risk, it can be found that the evolution of hoisting safety risk can be divided into four stages. The first stage (0–20 s) has the highest risk probability. In the second stage (20–40 s), the risk probability decreases rapidly. In the third stage (40–100 s), the risk probability tends to be flat. In the fourth stage (100–120 s), the risk probability increases rapidly.

The actual hoisting condition is also consistent with the calculated space-time evolution law. In the construction process of this case, there were two controllable safety risks during the hoisting process (neither caused property losses nor casualties). One is the acceleration lifting stage. As the lifting speed is too fast, the hook at the connection between the composite plate and the sling falls off. At this time, the laminated plate was about 20 cm away from the ground, and except for local damage of the laminated plate, no other safety loss was caused. Another is during the installation stage of the composite plate. Due to the fast-moving speed of the composite plate, small cracks appeared at the joints of the composite plate. No other safety losses were caused. Six-point lifting method was used in both cases. The size of the laminated plate is 180 × 3320 × 60, and the concrete grade is C30.

5.4. Virtual Model Construction and Running

Firstly, the BIM model is processed. This study converts the BIM model into IFC format files, then the format files are parsed and processed in JavaScript environment. The BIM model was carried on the web page using WebGL (as shown in Figure 11) after the lightweight of the BIM model.

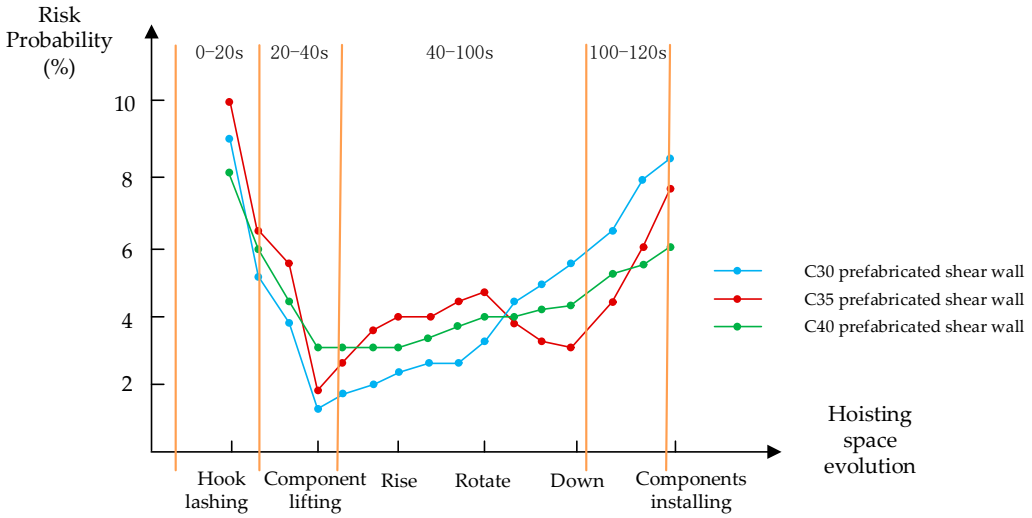


Figure 11. Temporal and spatial evolution law of safety risk in hoisting process.

Second, link the information to the database. The original information of sensors in the public database is carried in the BIM model. The location of the sensor and the real-time data collected can be viewed on the webpage. At the same time, the virtual model can simulate the hoisting process of the physical construction site according to the information of the positioning sensor. This simulation has a correspondence in time and space. This upgrades the traditional 3D BIM model to a 4D model based on time dimension under Digital Twin.

Finally, feedback control is carried out. When there is a safety risk in the hoisting forecast, the system will issue an alarm. The location and time of risks will be displayed on the web page of the display terminal of the manager, as well as the space-time evolution law of risks. In the construction site, the hoisting risk diagnosis report will be sent to the operator's display terminal through APP. Warning sounds and flashes will be issued at the corresponding locations on the scene. As shown in Figure 12, problems and management suggestions are marked on the web page.

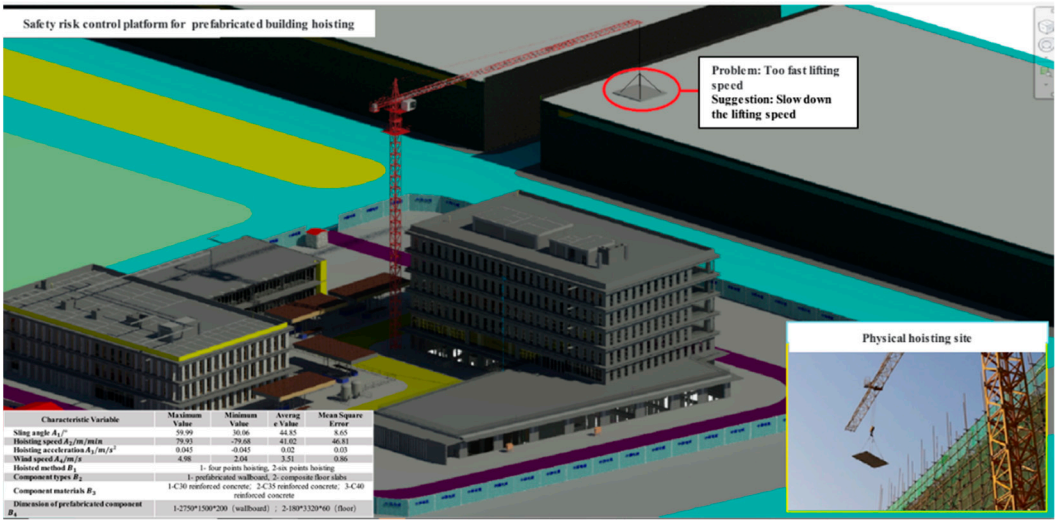


Figure 12. Safety risk control platform for prefabricated building hoisting.

6. Conclusions and Future Works

The current deficiency of prediction methods and data processing tools leads to low-level prefabricated construction hoisting safety management. In this paper, a safety risk management method is proposed to predict real-time risk and deduce risk evolution law

in prefabricated construction hoisting. First, the Digital Twin framework of the construction field is set up. Then, based on this framework, a Digital Twin risk prediction model is built for prefabricated construction hoisting. The model uses the IoT to collect and transmit components hoisting information and uses DT-SVM algorithm to analyze the data. The DT-SVM algorithm is an improvement of the traditional SVM algorithm, which can cooperate with the proposed Digital Twin model preferably. In addition, a large prefabricated project case is utilized to verify the feasibility of the proposed method.

The final results obtained from the framework of the case show that: 0–20 s, composite plate was in the accelerated lifting stage. At this time, the composite plate was located on the ground or within 5 m above the ground, and the stress of sling and composite plate was at a high level. There was a high probability of risk. At 20–40 s, composite plate was in a stable rising stage. The stress of sling and composite plate was at the equilibrium level, and the risk probability drops sharply. At 40–100 s, composite plate was in the stage of rotation and descent. At this time, the rotational stability in the horizontal direction decreased, but it was still at a relatively stable level. The probability of hoisting risk was raised gently. At 100–120 s, composite plate was in the installation stage. At this stage, the tower crane and the installation workers need to coordinate and adjust the position of the composite plate. The stability of composite plate decreased greatly, and the stress changed constantly, which led to the increase in risk probability.

This method has the following three important meanings:

1. This framework provides an efficient data interaction method, which can map multi-dimensional information on site to virtual model in time. Meanwhile, the framework can quickly feedback the simulation and calculation results of virtual space to managers and operators, and improve the level of safety management.
2. In this method, a DT-SVM algorithm which can interact with the Digital Twin framework is proposed. DT-SVM algorithm can process multi-source heterogeneous data to form unified sample data. It makes up for the shortage of the traditional SVM method in small sample data processing and provides a method for the fusion of prediction algorithm and information framework.
3. The framework applies Digital Twin to the safety risk prediction of prefabricated component hoisting, which effectively solves the defects of high hoisting risk, difficult prediction and low intelligence degree. This method also has vigorous application prospect in other construction process safety risk prediction.

This method provides a relatively accurate safety risk prediction method for pre-fabricated construction hoisting. It can improve the safety management level of pre-fabricated construction hoisting. However, there are still many deficiencies that need further research of this method:

1. The data in this paper are expanded by simulation on the basis of field data. Only to verify the feasibility of the framework, the space-time evolution law obtained is not applicable to all projects.
2. In the selection of characteristic variables, quantitative methods can also be used.
3. The existing methods have shortcomings in the aspect of automatic control. The risk problem also depends on the manual processing, which has not formed the automatic control mode. In the next step, the control method can be studied to improve the level of Digital Twin and carry it out in a manner with more "self-control".

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